

01304

DATA SET

DS1123

24. Where to Shop? Do you shop at the grocery store closest to home or do you look for the store that has the best prices? We compared the regular prices at four different grocery stores for eight items purchased on the same day.

Items	Stores			
	Vons	Ralphs	Stater Bros	WinCo
Salad mix, 360 g bag	3.99	2.79	1.99	1.78
Hillshire Farm® Beef Smoked Sausage, 420 g	4.29	4.29	3.99	2.50
Kellogg's Raisin Bran®, 750 g	4.49	5.49	4.49	3.15
Kraft® Philadelphia® Cream Cheese, 250 g	2.99	3.19	2.79	1.48
Kraft® Ranch Dressing, 450 mL	3.19	3.49	3.49	1.48
Treetop® Apple Juice, 2 L	2.99	3.49	3.49	1.58
Dial® Bar Soap, Gold, 125 g	5.99	6.49	5.79	5.14
Jif® Peanut Butter, Creamy, 850 g	5.15	5.49	4.79	4.34

- What are the blocks and treatments in this experiment?
- Do the data provide evidence to indicate that there are significant differences in prices from store to store? Support your answer statistically using the ANOVA printout that follows.
- Are there significant differences from block to block? Was blocking effective?

ANOVA: Price versus Item, Store

Analysis of Variance for Price

Source	DF	SS	MS	F	P
Store	3	13.192	4.3974	26.52	0.000
Item	7	40.995	5.8564	35.32	0.000
Error	21	3.482	0.1658		
Total	31	57.669			

Model Summary

S	R-sq	R-sq(adj)
0.407181	93.96%	91.09%

a. treatment: stores
blocks: items

b. $H_0: \mu_1 = \mu_2 = \mu_3$ H_a : at least one difference

$$\alpha = 0.05$$

$$F_{STAT} = \frac{MST}{MSE} \sim F_{(k-1, n-k)}$$

$$F^* = 30.52$$

$$RR: \{F: |F| > F_{(3, 21)}\} = \{F: |F| > 3.07\}$$

$$30.52 > 3.07 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ there is sufficient evidence to say that there is difference in price from store to store

c. $H_0: \mu_1 = \mu_2 = \mu_3$ H_a : at least one different

$$\alpha = 0.05$$

$$F_{STAT} = \frac{MST}{MSE} \sim F_{(k-1, n-k)}$$

$$F^* = 35.32$$

$$RR: \{F: |F| > F_{(8, 21)}\} = \{F: |F| > 2.4876\}$$

$$35.32 > 2.4876 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ We have sufficient evidence to say there are significant difference from block to block

$\Rightarrow$ block are effective